## Exercises and Problems for Section 17.2

### Exercises

In Exercises 1–6, find the velocity and acceleration vectors.

1. $x = 2 + 3t, y = 4 + t, z = 1 - t$
2. $x = 2 + 3t^2, y = 4 + t^2, z = 1 - t^2$
3. $x = t, y = t^2, z = t^3$
4. $x = t, y = t^3 - t$
5. $x = 3\cos t, y = 4\sin t$
6. $x = 3\cos(t^2), y = 3\sin(t^2), z = t^2$

In Exercises 7–12, find the velocity $\vec{v}(t)$ and speed $\|\vec{v}(t)\|$. Find any times at which the particle stops.

7. $x = t, y = t^2, z = t^3$
8. $x = \cos 3t, y = \sin 5t$
9. $x = 3t^2, y = t^3 + 1$
10. $x = (t-1)^2, y = 2, z = 2t^3 - 3t^2$
11. $x = 3\sin(t^2) - 1, y = 3\cos(t^2)$
12. $x = 3\sin^2 t, y = \cos t - 1, \quad z = t^2$

In Exercises 13–16, find the length of the curve.

13. $x = 3 + 5t, y = 1 + 4t, z = 3 - t$ for $1 \le t \le 2$. Check by calculating the length by another method.
14. $x = \cos 3t, y = \sin 5t$ for $0 \le t \le 2\pi$.
15. $x = \cos(e^t), y = \sin(e^t)$ for $0 \le t \le 1$. Check by calculating the length by another method.
16. $\vec{r}(t) = 2t\vec{i} + \ln t\vec{j} + t^2\vec{k}$ for $1 \le t \le 2$.

In Exercises 17–18, find the velocity and acceleration vectors of the uniform circular motion and check that they are perpendicular. Check that the speed and magnitude of the acceleration are constant.

17. $x = 3\cos(2\pi t), y = 3\sin(2\pi t), z = 0$
18. $x = 2\pi, y = 2\sin(3t), z = 2\cos(3t)$

In Exercises 19–20, find the velocity and acceleration vectors of the straight-line motion. Check that the acceleration vector points in the same direction as the velocity vector if the speed is increasing and in the opposite direction if the speed is decreasing.

19. $x = 2 + t^2, y = 3 - 2t^2, z = 5 - t^2$
20. $x = -2t^3 - 3t + 1, y = 4t^3 + 6t - 5, z = 6t^3 + 9t - 2$
21. Find parametric equations for the tangent line at $t = 2$ for Exercise 10.

### Problems

22. A particle passes through the point $P = (5, 4, -2)$ at time $t = 4$, moving with constant velocity $\vec{v} = 2\vec{i} - 3\vec{j} + \vec{k}$. Find a parametric equation for its motion.

In Problems 23–24, find all values of $t$ for which the particle is moving parallel to the $x$-axis and to the $y$-axis. Determine the end behavior and graph the particle's path.

23. $x = t^2 - 6t, \quad y = t^3 - 3t$
24. $x = t^3 - 12t, \quad y = t^2 + 10t$
25. The table gives $x$ and $y$ coordinates of a particle in the plane at time $t$. Assuming that the particle moves smoothly and that the points given show all the major features of the motion, estimate the following quantities:
    - **(a)** The velocity vector and speed at time $t = 2$.
    - **(b)** Any times when the particle is moving parallel to the $y$-axis.
    - **(c)** Any times when the particle has come to a stop.

| $t$ | 0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 |
|---|---|---|---|---|---|---|---|---|---|
| $x$ | 1 | 4 | 6 | 7 | 6 | 3 | 2 | 3 | 5 |
| $y$ | 3 | 2 | 3 | 5 | 8 | 10 | 11 | 10 | 9 |

26. A particle starts at the point $P = (3, 2, -5)$ and moves along a straight line toward $Q = (5, 7, -2)$ at a speed of 5 cm/sec. Let $x, y, z$ be measured in centimeters.
    - **(a)** Find the particle's velocity vector.
    - **(b)** Find parametric equations for the particle's motion.
27. A particle moves at a constant speed along a line from the point $P = (2, -1, 5)$ at time $t = 0$ to the point $Q = (5, 3, -1)$. Find parametric equations for the particle's motion if:
    - **(a)** The particle takes 5 seconds to move from $P$ to $Q$.
    - **(b)** The speed of the particle is 5 units per second.
28. A particle travels along the line $x = 1 + t, \ y = 5 + 2t, \ z = -7 + t$, where $t$ is in seconds and $x, y, z$ are in meters.
    - **(a)** When and where does the particle hit the plane $x + y + z = 1$?
    - **(b)** How fast is the particle going when it hits the plane? Give units.
29. A stone is thrown from a rooftop at time $t = 0$ seconds. Its position at time $t$ is given by

$$\vec{r}(t) = 10t\vec{i} - 5t\vec{j} + (6.4 - 4.9t^2)\vec{k}.$$

The origin is at the base of the building, which is standing on flat ground. Distance is measured in meters. The vector $\vec{i}$ points east, $\vec{j}$ points north, and $\vec{k}$ points up.

**(a)** How high is the rooftop above the ground?
**(b)** At what time does the stone hit the ground?
**(c)** How fast is the stone moving when it hits the ground?
**(d)** Where does the stone hit the ground?
**(e)** What is the stone's acceleration when it hits the ground?

**30.** A child wanders slowly down a circular staircase from the top of a tower. With $x, y, z$ in feet and the origin at the base of the tower, her position $t$ minutes from the start is given by

$$x = 10\cos t, \quad y = 10\sin t, \quad z = 90 - 5t.$$

**(a)** How tall is the tower?
**(b)** When does the child reach the bottom?
**(c)** What is her speed at time $t$?
**(d)** What is her acceleration at time $t$?

**31.** The origin is on flat ground and the $z$-axis points upward. For time $0 \le t \le 10$ in seconds and distance in centimeters, a particle moves along a path given by

$$\vec{r} = 2t\vec{i} + 3t\vec{j} + (100 - (t-5)^2)\vec{k}\,.$$

**(a)** When is the particle at the highest point? What is that point?
**(b)** When in the interval $0 \le t \le 10$ is the particle moving fastest? What is its speed at that moment?
**(c)** When in the interval $0 \le t \le 10$ is the particle moving slowest? What is its speed at that moment?

**32.** The function $w = f(x, y, z)$ has $\text{grad}\, f(7, 2, 5) = 4\vec{i} - 3\vec{j} + \vec{k}$. A particle moves along the curve $\vec{r}\,(t)$ arriving at the point $(7, 2, 5)$ with velocity $2\vec{i} + 3\vec{j} + 6\vec{k}$ when $t = 0$. Find the rate of change of $w$ with respect to time at $t = 0$.

**33.** Suppose $x$ measures horizontal distance in meters, and $y$ measures distance above the ground in meters. At time $t = 0$ in seconds, a projectile starts from a point $h$ meters above the origin with speed $v$ meters/sec at an angle $\theta$ to the horizontal. Its path is given by

$$x = (v\cos\theta)t, \quad y = h + (v\sin\theta)t - \frac{1}{2}gt^2.$$

Using this information about a general projectile, analyze the motion of a ball which travels along the path

$$x = 20t, \quad y = 2 + 25t - 4.9t^2.$$

**(a)** When does the ball hit the ground?
**(b)** Where does the ball hit the ground?
**(c)** At what height above the ground does the ball start?
**(d)** What is the value of $g$, the acceleration due to gravity?
**(e)** What are the values of $v$ and $\theta$?

**34.** A particle is moving on a path in the $xz$-plane given by $x = 20t$, $z = 5t - 0.5t^2$, where $z$ is the height of the particle above the ground in meters, $x$ is the horizontal distance in meters, and $t$ is time in seconds.

**(a)** What is the equation of the path in terms of $x$ and $z$ only?
**(b)** When is the particle at ground level?
**(c)** What is the velocity of the particle at time $t$?
**(d)** What is the speed of the particle at time $t$?
**(e)** Is the speed ever 0?
**(f)** When is the particle at the highest point?

**35.** The base of a 20-meter tower is at the origin; the base of a 20-meter tree is at $(0, 20, 0)$. The ground is flat and the $z$-axis points upward. The following parametric equations describe the motion of six projectiles each launched at time $t = 0$ in seconds.

(I) $\vec{r}\,(t) = (20 + t^2)\vec{k}$
(II) $\vec{r}\,(t) = 2t^2\vec{j} + 2t^2\vec{k}$
(III) $\vec{r}\,(t) = 20\vec{i} + 20\vec{j} + (20 - t^2)\vec{k}$
(IV) $\vec{r}\,(t) = 2t\vec{j} + (20 - t^2)\vec{k}$
(V) $\vec{r}\,(t) = (20 - 2t)\vec{i} + 2t\vec{j} + (20 - t)\vec{k}$
(VI) $\vec{r}\,(t) = t\vec{i} + t\vec{j} + t\vec{k}$

**(a)** Which projectile is launched from the top of the tower and goes downward? When and where does it hit the ground?
**(b)** Which projectile hits the top of the tree? When? From where is it launched?
**(c)** Which projectile is not launched from somewhere on the tower and hits the tree? Where and when does it hit the tree?

**36.** A particle moves on a circle of radius 5 cm, centered at the origin, in the $xy$-plane ($x$ and $y$ measured in centimeters). It starts at the point $(0, 5)$ and moves counterclockwise, going once around the circle in 8 seconds.

**(a)** Write a parameterization for the particle's motion.
**(b)** What is the particle's speed? Give units.

**37.** Determine the position vector $\vec{r}\,(t)$ for a rocket which is launched from the origin at time $t = 0$ seconds, reaches its highest point of $(x, y, z) = (1000, 3000, 10{,}000)$, where $x, y, z$ are in meters, and after the launch is subject only to the acceleration due to gravity, 9.8 m/sec$^2$.

**38.** Emily is standing on the outer edge of a merry-go-round, 10 meters from the center. The merry-go-round completes one full revolution every 20 seconds. As Emily passes over a point $P$ on the ground, she drops a ball from 3 meters above the ground.

**(a)** How fast is Emily going?
**(b)** How far from $P$ does the ball hit the ground? (The acceleration due to gravity is 9.8 m/sec$^2$.)
**(c)** How far from Emily does the ball hit the ground?

**39.** A point $P$ moves in a circle of radius $a$. Show that $\vec{r}\,(t)$, the position vector of $P$, and its velocity vector $\vec{r}\,'(t)$ are perpendicular.

**40.** A wheel of radius 1 meter rests on the $x$-axis with its center on the $y$-axis. There is a spot on the rim at the point $(1, 1)$. See Figure 17.15. At time $t = 0$ the wheel starts rolling on the $x$-axis in the direction shown at a rate of 1 radian per second.

**(a)** Find parametric equations describing the motion of the center of the wheel.
**(b)** Find parametric equations describing the motion of the spot on the rim. Plot its path.

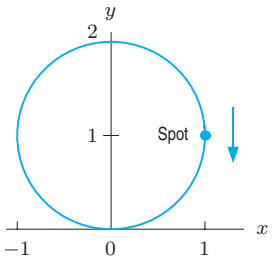



Figure 17.15

**41.** Suppose $\vec{r}\,(t) = \cos t\,\vec{i} + \sin t\,\vec{j} + 2t\,\vec{k}$ represents the position of a particle on a helix, where $z$ is the height of the particle above the ground.

**(a)** Is the particle ever moving downward? When?
**(b)** When does the particle reach a point 10 units above the ground?
**(c)** What is the velocity of the particle when it is 10 units above the ground?
**(d)** When it is 10 units above the ground, the particle leaves the helix and moves along the tangent. Find parametric equations for this tangent line.

**42.** Show that the helix $\vec{r} = \alpha\cos t\,\vec{i} + \alpha\sin t\,\vec{j} + \beta t\vec{k}$ is parameterized with constant speed.

**43.** An ant crawls along the radius from the center to the edge of a circular disk of radius 1 meter, moving at a constant rate of 1 cm/sec. Meanwhile, the disk is turning counterclockwise about its center at 1 revolution per second.

**(a)** Parameterize the motion of the ant.
**(b)** Find the velocity and speed of the ant.
**(c)** Determine the acceleration and magnitude of the acceleration of the ant.

**44.** The motion of a particle is given by $\vec{r}\,(t) = R\cos(\omega t)\vec{i} + R\sin(\omega t)\vec{j}$, with $R > 0, \omega > 0$.

**(a)** Show that the particle moves on a circle and find the radius, direction, and period.
**(b)** Determine the velocity vector of the particle and its direction and speed.
**(c)** What are the direction and magnitude of the acceleration vector of the particle?

**45.** You bicycle along a straight flat road with a safety light attached to one foot. Your bike moves at a speed of 25 km/hr and your foot moves in a circle of radius 20 cm centered 30 cm above the ground, making one revolution per second.

**(a)** Find parametric equations for $x$ and $y$ which describe the path traced out by the light, where $y$ is distance (in cm) above the ground.
**(b)** Sketch the light's path.
**(c)** How fast (in revolutions/sec) would your foot have to be rotating if an observer standing at the side of the road sees the light moving backward?

**46.** How do the motions of objects $A$ and $B$ differ, if $A$ has position vector $\vec{r}_{\,A}(t)$ and $B$ has position vector $\vec{r}_{\,B}(t) = \vec{r}_{\,A}(2t)$ for $t \geq 0$. Illustrate your answer with $\vec{r}_{\,A}(t) = t\vec{i} + t^2\vec{j}$.

**47.** At time $t = 0$ an object is moving with velocity vector $\vec{v} = 2\vec{i} + \vec{j}$ and acceleration vector $\vec{a} = \vec{i} + \vec{j}$. Can it be in uniform circular motion about some point in the plane?

**48.** Figure 17.16 shows the velocity and acceleration vectors of an object in uniform circular motion about a point in the plane at a particular moment. Is it moving round the circle in the clockwise or counterclockwise direction?

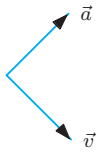



Figure 17.16

**49.** The position of a particle at time $t$ is given by $\vec{r}\,(t)$. Let $r = \|\vec{r}\,\|$ and $\vec{a}$ be a constant vector. Differentiate:

**(a)** $\vec{r}\cdot\vec{r}$ **(b)** $\vec{a}\times\vec{r}$ **(c)** $r^3\vec{r}$

**50.** The function $f(x, y, z)$ is defined and smooth at every point in 3-space and grad $f(1, 7, 2) = \vec{i} - (\sqrt{6})\vec{j} + \vec{k}$. The curve $C$ is $\vec{r} = (t+1)^2\vec{i} + 7\cos t\,\vec{j} + 2e^t\vec{k}$.

**(a)** Find an equation of the tangent plane to the level surface of $f$ at the point $(1, 7, 2)$.
**(b)** Find the angle between the normal to the level surface of $f$ and the tangent to the curve $C$ at $(1, 7, 2)$. (Note: There are two possible angles; give the smaller one. Your answer should be in radians.)
**(c)** With $x, y, z$ in centimeters, let $f$ be the concentration of a pollutant in parts per million (ppm) at the point $(x, y, z)$. A particle moves along the curve $C$ with the given parameterization and $t$ in seconds. Find how fast the concentration is changing at the time $t = 0$. Give units with your answer.

**51.** Let $\vec{v}\,(t)$ be the velocity of a particle moving in the plane. Let $s(t)$ be the magnitude of $\vec{v}$ and let $\theta(t)$ be the angle of $\vec{v}\,(t)$ with the positive $x$-axis at time $t$, so that $\vec{v} = s\cos\theta\,\vec{i} + s\sin\theta\,\vec{j}$.

Let $\vec{T}$ be the unit vector in the direction of $\vec{v}$, and let $\vec{N}$ be the unit vector in the direction of $\vec{k} \times \vec{v}$, perpendicular to $\vec{v}$. Show that the acceleration $\vec{a}\,(t)$ is given by

$$\vec{a} = \frac{ds}{dt}\vec{T} + s\frac{d\theta}{dt}\vec{N}.$$

This shows how to separate the acceleration into the sum of one component, $\frac{ds}{dt}\vec{T}$, due to changing speed and a perpendicular component, $s\frac{d\theta}{dt}\vec{N}$, due to changing direction of the motion.

**52.** A point particle $P$ is acted on by a force, $\vec{F}$, which is directed toward a fixed point $O$; this is called a central force. Let $\vec{r}\,(t)$ be the position of the particle with respect to $O$ and let $\vec{v}\,(t)$ be its velocity. Use Newton's second law $\vec{F} = m\vec{a}$, where $m$ is mass and $\vec{a}$ is acceleration, to show that $\vec{r}\,(t) \times \vec{v}\,(t) = \vec{c}$, a constant vector. Explain why this tells us that the particle always moves in the same plane.

**53.** In this problem we justify the formula for the length of a curve given on page 932. Suppose the curve $C$ is given by smooth parametric equations $x = x(t)$, $y = y(t)$, $z = z(t)$ for $a \le t \le b$. By dividing the parameter interval $a \le t \le b$ at points $t_1, \ldots, t_{n-1}$ into small segments of length $\Delta t = t_{i+1} - t_i$, we get a corresponding division of the curve $C$ into small pieces. See Figure 17.17, where the points $P_i = (x(t_i), y(t_i), z(t_i))$ on the curve $C$ correspond to parameter values $t = t_i$. Let $C_i$ be the portion of the curve $C$ between $P_i$ and $P_{i+1}$.

**(a)** Use local linearity to show that

$$\text{Length of } C_i \approx \sqrt{x'(t_i)^2 + y'(t_i)^2 + z'(t_i)^2}\,\Delta t.$$

**(b)** Use part (a) and a Riemann sum to explain why

$$\text{Length of } C = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt.$$

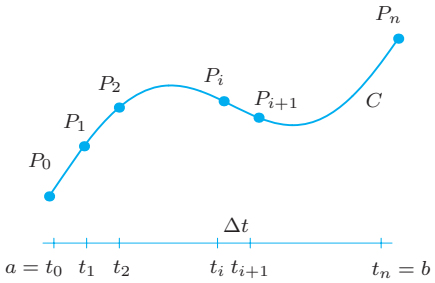



**Figure 17.17**: A subdivision of the parameter interval and the corresponding subdivision of the curve $C$

## Strengthen Your Understanding

In Problems 54–56, explain what is wrong with the statement.

**54.** When a particle moves around a circle its velocity and acceleration are always orthogonal.

**55.** A particle with position $\vec{r}\,(t)$ at time $t$ has acceleration equal to 3 m/sec$^2$ at time $t = 0$.

**56.** A parameterized curve $\vec{r}\,(t)$, $A \le t \le B$, has length $B - A$.

In Problems 57–58, give an example of:

**57.** A function $\vec{r}\,(t)$ such that the particle with position $\vec{r}\,(t)$ at time $t$ has velocity $\vec{v} = \vec{i} + 2\vec{j}$ and acceleration $\vec{a} = 4\vec{i} + 6\vec{k}$ at $t = 0$.

**58.** An interval $a \le t \le b$ corresponding to a piece of the helix $\vec{r}\,(t) = \cos t\vec{i} + \sin t\vec{j} + t\vec{k}$ of length 10.

Are the statements in Problems 59–66 true or false? Give reasons for your answer.

**59.** A particle whose motion in the plane is given by $\vec{r}\,(t) = t^2\vec{i} + (1-t)\vec{j}$ has the same velocity at $t = 1$ and $t = -1$.

**60.** A particle whose motion in the plane is given by $\vec{r}\,(t) = t^2\vec{i} + (1-t)\vec{j}$ has the same speed at $t = 1$ and $t = -1$.

**61.** If a particle is moving along a parameterized curve $\vec{r}\,(t)$ then the acceleration vector at any point is always perpendicular to the velocity vector at that point.

**62.** If a particle is moving along a parameterized curve $\vec{r}\,(t)$ then the acceleration vector at a point cannot be parallel to the velocity vector at that point.

**63.** If $\vec{r}\,(t)$ for $a \le t \le b$ is a parameterized curve, then $\vec{r}\,(-t)$ for $a \le t \le b$ is the same curve traced backward.

**64.** If $\vec{r}\,(t)$ for $a \le t \le b$ is a parameterized curve $C$ and the speed $||\vec{v}\,(t)|| = 1$, then the length of $C$ is $b - a$.

**65.** If a particle moves with motion $\vec{r}\,(t) = 3t\vec{i} + 2t\vec{j} + t\vec{k}$, then the particle stops at the origin.

**66.** If a particle moves with constant speed, the path of the particle must be a line.

For Problems 67–70, decide if the statement is true or false for all smooth parameterized curves $\vec{r}\,(t)$ and all values of $t$ for which $\vec{r}\,'(t) \neq \vec{0}$.

**67.** The vector $\vec{r}\,'(t)$ is tangent to the curve at the point with position vector $\vec{r}\,(t)$.

**68.** $\vec{r}\,'(t) \times \vec{r}\,(t) = \vec{0}$

**69.** $\vec{r}\,'(t) \cdot \vec{r}\,(t) = 0$

**70.** $\vec{r}\,''(t) = -\omega^2\vec{r}\,(t)$